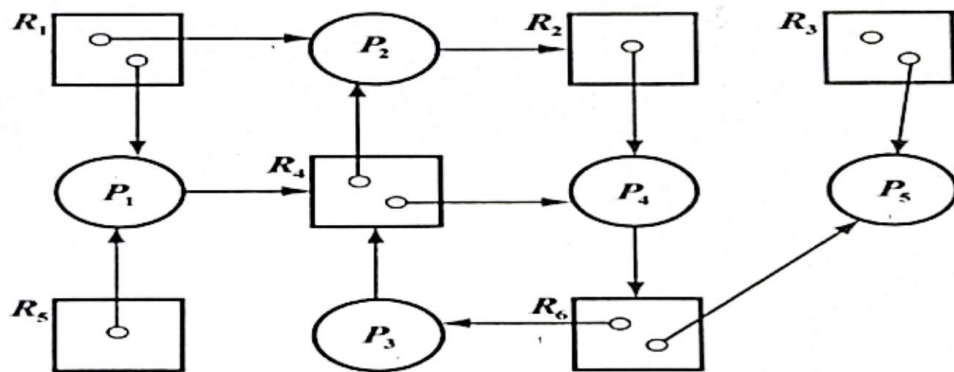


1. Assume that there are 5 processes. Consider the following data table and formulate banker algorithm to find out the safe sequence. Here, Max Instances of Resource Type A = 3, Max Instances of Resource Type B = 17, Max Instances of Resource Type C = 16, Max Instances of Resource Type D = 12

Process	Allocation Matrix				Max. Need Matrix			
	A	B	C	D	A	B	C	D
P1	0	1	1	0	0	2	1	0
P2	1	2	3	1	1	6	5	2
P3	1	3	6	5	2	3	6	6
P4	0	6	3	2	0	6	5	2
P5	0	0	1	4	0	6	5	6

2. Conclude whether there is a deadlock in this RAG(Resource Allocation Graph) or not.



Sagnik Saha Dibbay

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Q-1

Process		Max Allocation				Max Need				Need			
		A	B	C	D	A	B	C	D	A	B	C	D
P ₁	1	0	1	1	0	0	2	1	0	0	1	0	0
P ₂	0	1	2	3	1	1	6	5	2	0	4	2	1
P ₃	1	1	3	6	5	2	3	6	6	1	0	0	1
P ₄	0	0	6	3	2	0	6	5	2	0	0	2	0
P ₅	0	0	0	1	4	0	6	5	6	0	6	4	2
		2	12	14	12	1	0	0	1	0	0		

Max-Allocation (0,0,0,0,0,0) ← available init

A B C D

3 17 16 12

2 12 14 12

1 1 5 2 0

0 1 1 0 ⇒ P₁

1 6 3 0

0 6 3 2 ⇒ P₄

1 12 6 2

1 2 3 1 ⇒ P₂

2 14 9 3

1 3 6 5 ⇒ P₃

3 17 15 8

0 0 1 4 = P₅

3 17 16 12

Available

0 0 0 0 0 0

Safe sequence -

1 0 0 1 0 0

P₁ → P₄ → P₂ → P₃ → P₅

1 0

(Ans)

1-2

Initial available $\rightarrow (0, 0, 0, 0, 0, 0)$ of total 11A-x0M

Process	<u>Available</u>	<u>R₁</u>	<u>R₂</u>	<u>R₃</u>	<u>R₄</u>	<u>R₅</u>	<u>R₆</u>
P ₀	5	1	0	1	0	0	1
P ₄	0	0	1	1	1	0	1
P ₁	0	1	1	1	0	1	1
P ₂	0	0	0	1			
P ₃	5	0	0	0	0		
P ₅	5	0	0	1	1		
P ₆	1	0	0	0	1		
P ₇	0	0	0	0	1		
P ₈	0	0	0	0	0		
P ₉	0	0	0	0	0		

P.T.O

P.T.O

Step	Allocation						Process	Available					
	<u>R₁</u>	<u>R₂</u>	<u>R₃</u>	<u>R₄</u>	<u>R₅</u>	<u>R₆</u>		<u>R₁</u>	<u>R₂</u>	<u>R₃</u>	<u>R₄</u>	<u>R₅</u>	<u>R₆</u>
—	—	—	—	—	—	—	—	0	0	0	0	0	0
1	0	0	1	0	0	1	P5	0	0	1	0	0	1
2	0	1	0	1	0	0	P4	0	1	1	1	0	1
3	1	0	0	0	1	0	P1	1	1	1	1	1	1
4	1	0	0	1	0	0	P2	2	1	1	2	1	1
5	0	0	0	0	0	1	P3	2	1	1	2	1	2

Safe sequence →

P5 → P4 → P1 → P2 → P3

Ans

So the system is Deadlock free.